Magnetic Fields in Matter

When we say “magnetism”, we only think of a bar magnet or a needle magnet or a similar one, and we don’t think of moving charges. Yet all the magnetic phenomena are due to electric charges in motion.

If we look at piece of magnetic material on an atomic scale, we find tiny currents due to electrons revolving in orbits or spinning about their own axes. These current loops create magnetic dipoles. These dipoles cancel each other due to random orientation of atoms.

When a magnetic field is applied, a net alignment of these magnetic dipole occurs and the medium becomes magnetically polarized or magnetized.

Types of Magnetic Ordering

Those not having a permanent magnetic moment

They develop a magnetization Opposite to **B**

(diamagnetic)

Those having a permanent magnetic moment

They develop a magnetization Parallel to **B**

(paramagnetic)

Paramagnetics still have further classes :
Ferromagnetic, Antiferromagnetic, Ferrimagnetic

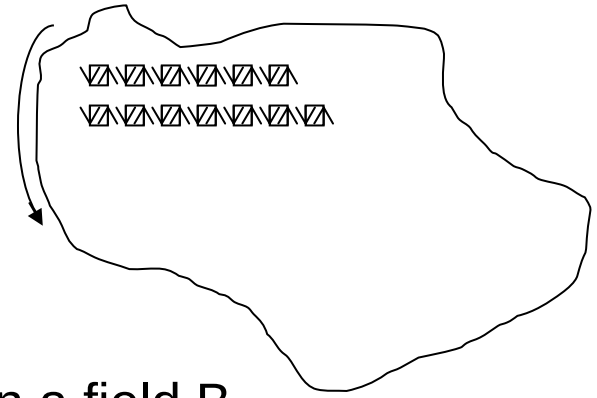
Diamagnetism - Present in every material.
Change in orbital speed of the electron - resulting in orbital dipole moment in a direction opposite to the field

Paramagnetism - The dipoles associated with the spins of unpaired electrons experience a torque
The dipoles tend to align along the direction of the external field **B**.

Ferromagnetism – Associated with the spins of unpaired electrons.
A strong exchange interaction between neighboring dipoles exist.
The neighboring dipoles tend to align in the same direction.
They exhibit a domain structure.
Within the domains the dipoles are aligned in the same direction.
The domains are randomly oriented.

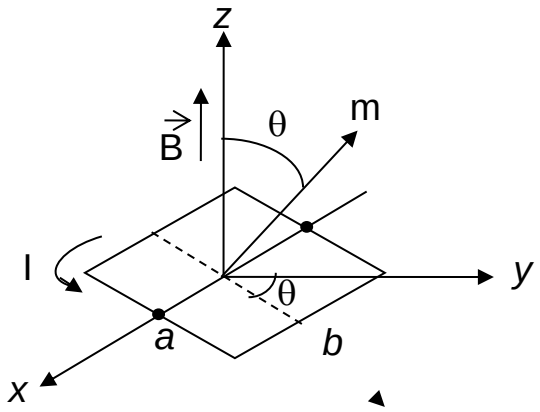
Influence of Magnetic Field on a dipole

Will experience a Torque



Consider a tiny rectangular (circular) current loop in a field B.
(Any current loop can be built-up from small loops like this with all internal sides canceling)

Let the magnetic field be in z-direction.
centre the loop at the origin and tilt it at an angle θ from the z-axis towards the y-axis.



Force on the two sloping sides and the horizontal sides are equal and opposite and cancel out

Force on a dipole

In a uniform field the net force on a current loop is zero.

$$\vec{F} = I\oint(\vec{dl}\times\vec{B}) = I\left(\oint\vec{dl}\right)\times\vec{B} = 0$$

B is constant, so comes out of integral
Net displacement ie., integral dl over a closed loop is zero

This is not the case when the field is non-uniform

In a non-uniform field, there will be a torque plus force

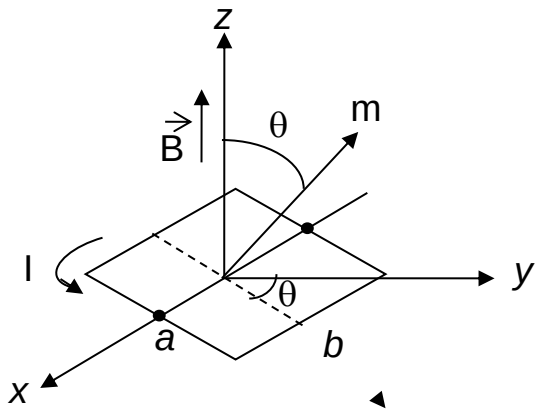
But they generate a Torque

$$\vec{N} = aF\sin\theta\,\hat{e}_x$$

$$F = IbB$$

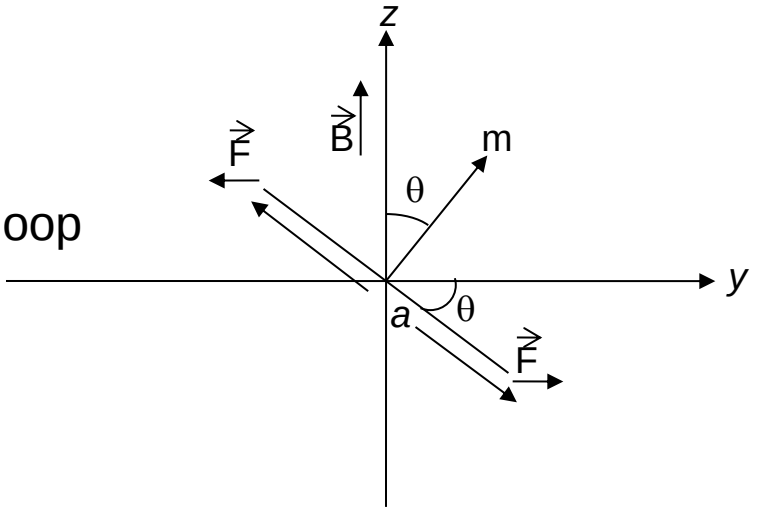
$$\vec{N} = IabB\sin\theta\,\hat{e}_x = mB\sin\theta\,\hat{e}_x$$

$$\vec{N} = \vec{m}\times\vec{B}$$



F is the force on each of the sloping segment

m = I a b is the magnetic dipole moment of the loop



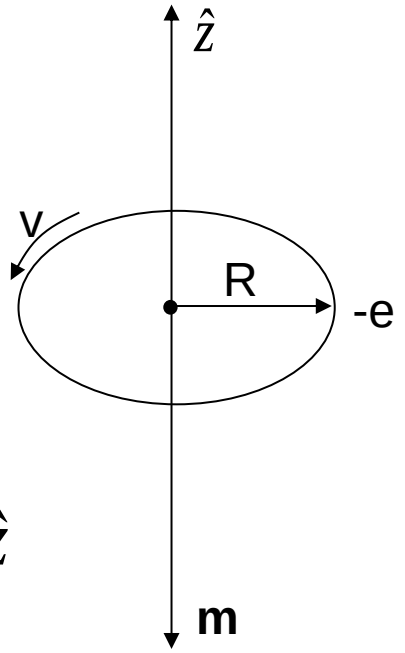
In electrostatics $\vec{N} = \vec{p}\times\vec{E}$

This is responsible for **paramagnetism**.
Torque due to electrons with opposite spins cancel out.
When we have odd number of electrons, then the unpaired electron is subjected to magnetic torque

Effect of a Magnetic field on Atomic orbitals

The orbit is a circle of radius R

period $T = \frac{2\pi R}{v}$ is so short that you can consider it as a steady current



$$I = \frac{e}{T} = \frac{ev}{2\pi R}$$

Orbital dipole moment is $\vec{m} = I(\pi R^2) = -\frac{1}{2} evR\hat{z}$

- ive sign account for the negative charge of the electron

What happens when this dipole is kept in a magnetic field?

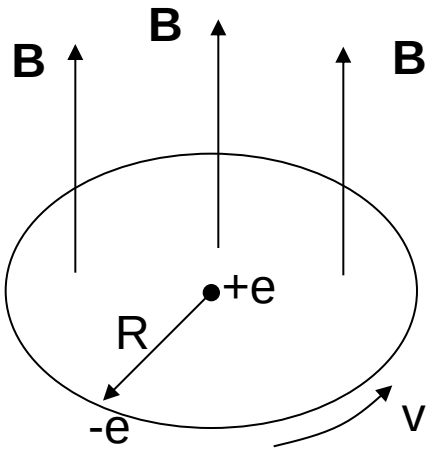
It is difficult to tilt orbital plane by the torque – so no contribution to paramagnetism
But, there is another significant effect on the orbital motion.
The electron speeds up or slow down, depending on the orientation of **B**

Before applying **B**, Centripetal acceleration is sustained by electrical forces alone

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{R^2} = m_e \frac{v^2}{R}$$

There is an additional force by magnetic field, when $\vec{B}$ is applied

$$-e\left(\vec{v}\times\vec{B}\right)$$



Let B be perpendicular to the plane of the orbit

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{R^2}+e\bar{v}B=m_e\frac{\bar{v}^2}{R}$$

The new speed $\bar{v}$ is greater than v

$$e\bar{v}B=\frac{m_e}{R}\left(\bar{v}^2-v^2\right)\quad=\frac{m_e}{R}\left(\bar{v}+v\right)\left(\bar{v}-v\right)$$

assuming $\Delta v=\bar{v}-v$ **to be very small**

$$\Delta v=\frac{eRB}{2m_e}$$

As **B** is applied, the electron speeds up

Change in orbital speed means a change in dipole moment

$$\Delta\vec{m} = -\frac{1}{2}e(\Delta v)R\hat{z} = -\frac{e^2R^2}{4m_e}\vec{B}$$

Change in **m** is opposite to the direction of **B**

An electron encircling the other way would have a dipole moment
Pointing upward, but such an orbit would be slowed down by the field
So the change is still opposite to the field

For the electrons to be accelerated or decelerated, some work is to be done.
Does the magnetic field do this ??

We said that magnetic fields do no work… are we contradicting ??

No we are not contradicting.. The change in magnetic field produces an Electric field which does this work

So, in the presence of a magnetic field, each atom picks up an "extra" dipole Moment and these are antiparallel to the field.
This mechanism is responsible for diamagnetism

This diamagnetism is there with all atoms, but typically much weaker than paramagnetism (which is caused due to spin).
Mainly observed in atoms with even number of electrons, where Paramagnetism is absent.

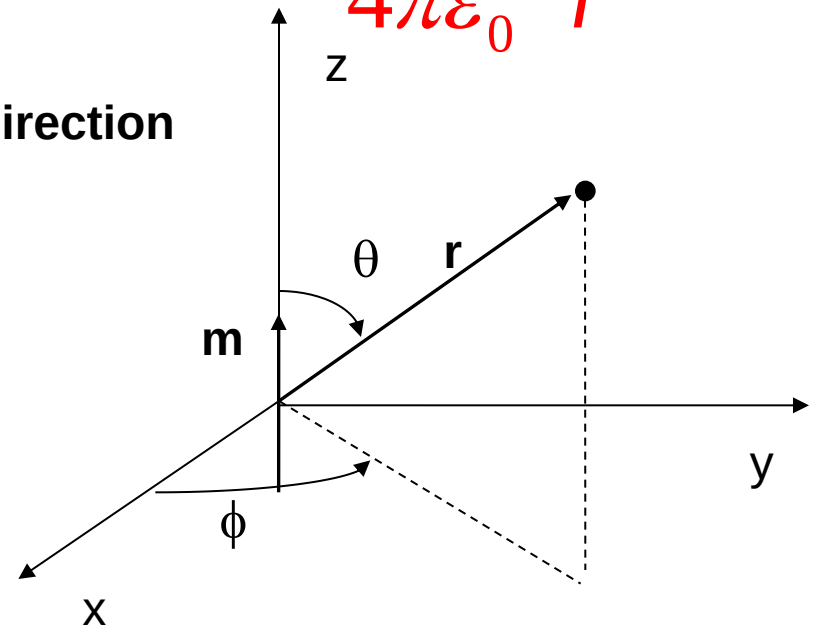
Magnetic dipole field – field due to magnetic dipole

RECAP

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\hat{r}}{r^2}$$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}.\hat{r}}{r^2}$$

Put m at the origin and let it point in the z-direction



$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{m\sin\theta}{r^2}\hat{e}_\phi$$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$$

Magnetic field is

$$\vec{B}_{dip}(\vec{r}) = \vec{\nabla}\times\vec{A} = \frac{\mu_0 m}{4\pi r^3}\left(2\cos\theta\hat{e}_r + \sin\theta\hat{e}_\theta\right)$$

$$\text{Field due to electric dipole} = \frac{p}{4\pi\varepsilon_0 r^3}\left(2\cos\theta\hat{e}_r + \sin\theta\hat{e}_\theta\right)$$

Magnetic field is

RECAP

$$\vec{B}_{dip}(\vec{r}) = \vec{\nabla} \times \vec{A} = \frac{\mu_0 m}{4\pi r^3}\left(2\cos\theta \hat{e}_r + \sin\theta \hat{e}_\theta\right)$$

$$\text{Field due to electric dipole} = \frac{p}{4\pi\varepsilon_0 r^3}\left(2\cos\theta \hat{e}_r + \sin\theta \hat{e}_\theta\right)$$

In general

$$B_{dip}\left(\vec{r}\right) = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}]$$

$$E_{dip}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{1}{r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right]$$

Magnetization

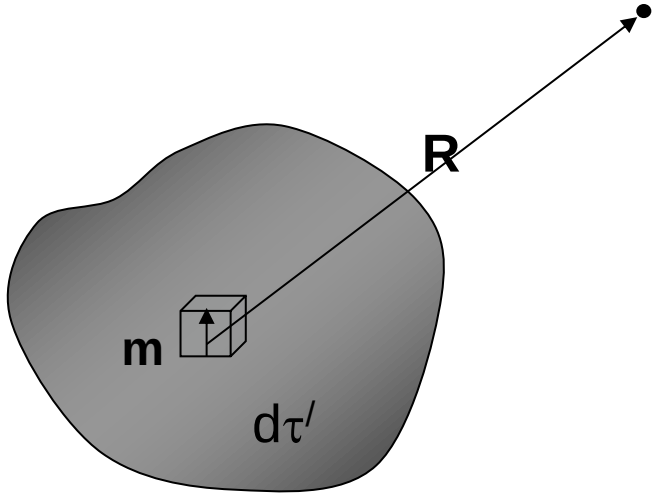
$\vec{M} =$ Magnetic dipole moment per unit volume is called magnetization

Paramagnet is attracted into the field The field inside is > field outside

Diamagnet is repelled The field inside is < field outside (sometimes zero)

Field due to a Magnetized object

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\hat{R}}{R^2}$$



Each volume element carries a dipole moment $\vec{M}d\tau'$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{M}(\vec{r}')\times\hat{R}}{R^2}d\tau'$$

$$\vec{\nabla}'\frac{1}{R} = \frac{\hat{R}}{R^2}$$

$$\vec{A} = \frac{\mu_0}{4\pi}\int\left[\vec{M}(\vec{r}')\times\left(\vec{\nabla}'\frac{1}{R}\right)\right]d\tau'$$

Integrating by parts and using product rule, we get

(i.e.) $$\nabla\times fA = f\left(\nabla\times A\right) - A\times\nabla f$$

$$\vec{A} = \frac{\mu_0}{4\pi}\left\{\int\frac{1}{R}\left[\vec{\nabla}'\times\vec{M}(\vec{r}')\right]d\tau' - \int\vec{\nabla}'\times\left[\frac{\vec{M}(\vec{r}')}{R}\right]d\tau'\right\}$$

Second term can be made into a surface integral as follows

The divergence theorem is, $\int \vec{\nabla} \cdot \vec{K} d\tau = \int \vec{K}.d\vec{a}$ Where **K** is some vector

Put $\vec{K} = \vec{v} \times \vec{c}$ **in the divergence theorem,** Where **c** is a constant vector

$$\int \vec{\nabla} \cdot (\vec{v} \times \vec{c}) d\tau = \int (\vec{v} \times \vec{c}).d\vec{a}$$

LHS can be written as

$$\vec{\nabla} \cdot (\vec{v} \times \vec{c}) = \vec{c} \cdot (\vec{\nabla} \times \vec{v}) - \vec{v} \cdot (\vec{\nabla} \times \vec{c}) = \vec{c} \cdot (\vec{\nabla} \times \vec{v})$$

since $\nabla \bullet (A \times B) = B \bullet (\nabla \times A) - A \bullet (\nabla \times B)$

RHS can be written as

$$d\vec{a} \cdot (\vec{v} \times \vec{c}) = \vec{c} \cdot (d\vec{a} \times \vec{v}) = -\vec{c} \cdot (\vec{v} \times d\vec{a})$$

since $A \bullet (B \times C) = B \bullet (C \times A) = C \bullet (A \times B)$

Therefore $\int \vec{c}.(\vec{\nabla} \times \vec{v}) d\tau = -\int \vec{c} \cdot (\vec{v} \times d\vec{a})$

Take **c** outside $\int (\vec{\nabla} \times \vec{v}) d\tau = -\int (\vec{v} \times d\vec{a})$

$$\vec{A} = \frac{\mu_0}{4\pi}\int \frac{1}{R}\left[\vec{\nabla}' \times \vec{M}(\vec{r}')\right] d\tau' + \frac{\mu_0}{4\pi}\oint \frac{1}{R}\left[\vec{M}(\vec{r}') \times d\vec{a}'\right]$$

$\sigma_b = -\vec{\nabla}\cdot\vec{P}$ $\quad$ $\sigma_b = \vec{P}\cdot\hat{n}$

$$\vec{J}_b = \vec{\nabla} \times \vec{M} \qquad \vec{K}_b = \vec{M} \times \hat{n}$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int_V \frac{\vec{J}_b(\vec{r}')}{R} d\tau' + \frac{\mu_0}{4\pi}\oint_S \frac{\vec{K}_b(\vec{r}')}{R} da'$$

$$\vec{J}_b = \vec{\nabla} \times \vec{M} \qquad \vec{K}_b = \vec{M} \times \hat{n}$$

Volume current $\qquad$ Surface current

are called the bound currents

$$\rho_b = -\vec{\nabla}\bullet\vec{P} \qquad \sigma_b = \vec{P}\bullet\hat{n}$$

Example - Uniformly magnetized sphere

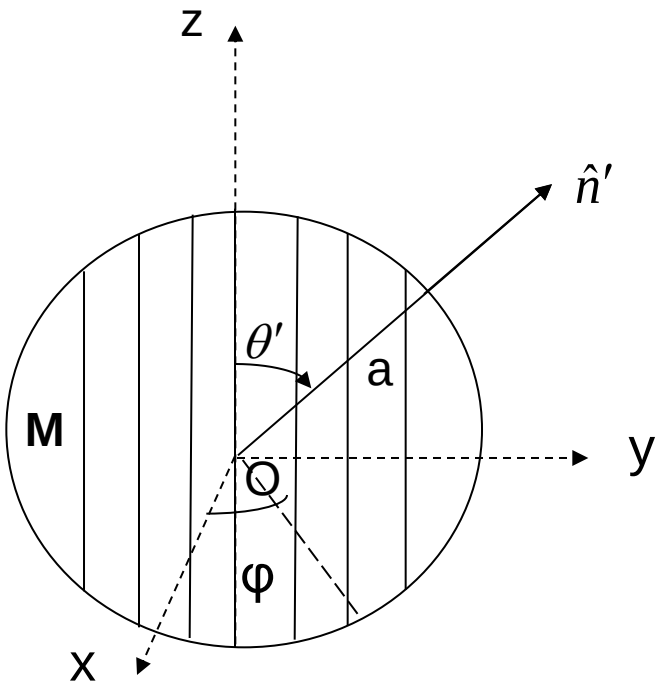
Let us choose Z-axis along direction of M

$\vec{M} = M\hat{z}$ uniform magnetization, hence

$$\vec{J}_b = \vec{\nabla} \times \vec{M} = 0$$

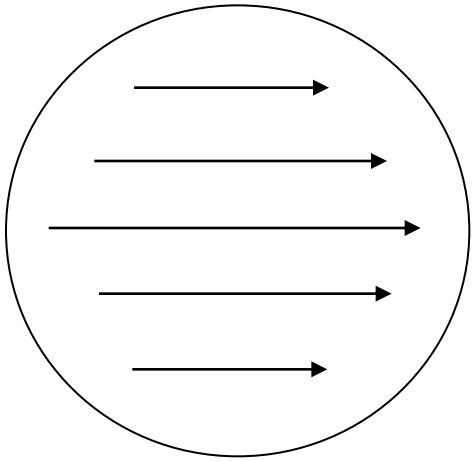
Here $\hat{n}' = \vec{r}\,'$ since it is a sphere

$$\vec{K}_b = \vec{M} \times \hat{n}' = M \sin\theta' \hat{e}_\phi$$



Surface currents are along the lines of "latitude" with a magnitude zero at the "poles" and a maximum at the equator

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_S \frac{\vec{K}_b(\vec{r}\,')}{R} da'$$



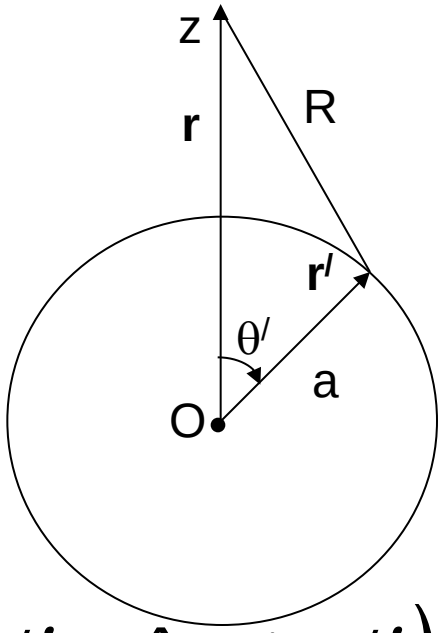
$$da' = a^2 \sin\theta' d\theta' d\phi' \qquad \vec{B} = \nabla \times \vec{A}$$

$$\vec{B}(\vec{r}) = \frac{\mu_0 M a^2}{4\pi} \int_0^{2\pi} \int_0^{\pi} \frac{\left(\hat{e}_{\phi'} \times \vec{R}\right) \sin^2 \theta' d\theta' d\phi'}{R^3}$$

$$\vec{r} = z\hat{e}_z \quad \vec{r}' = a\hat{e}_{r'}$$

$$\vec{R} = z\hat{e}_z - a\hat{e}_{r'}$$

$$R = \left(z^2 + a^2 - 2za\cos\theta'\right)^{1/2}$$



$$\hat{e}_{\phi'} \times \vec{R} = z\sin\theta'\hat{e}_{r'} + \left(z\cos\theta' - a\right)\hat{e}_{\theta'}$$

$$= a\sin\theta'\hat{e}_z + \left(z - a\cos\theta'\right)\left(\hat{e}_x \cos\phi' + \hat{e}_y \sin\phi'\right)$$

x and y components will vanish on integration

$$B_z(z) = \frac{\mu_0 M a^3}{4\pi} \int_0^{2\pi} \int_0^{\pi} \frac{\sin^3\theta' d\theta' d\phi'}{\left(z^2 + a^2 - 2za\cos\theta'\right)^{3/2}}$$

Integration over ϕ' gives a value 2π

We can use $\sin^2\theta' = 1 - \cos^2\theta'$ **and change it to a new variable** μ

$$B_z(z) = \frac{\mu_0 M a^3}{2} \int_1^{-1} \frac{\left(1-\mu^2\right) d\mu}{\left(z^2 + a^2 - 2za\mu\right)^{3/2}}$$

Value of the integral is

$$= \frac{2}{3z^3 a^3} \left\{ \left(z^2 + a^2\right) \left[|z+a| - |z-a| \right] - za \left[|z+a| + |z-a| \right] \right\}$$

Inside the sphere z < a, $|z-a| = a - z$

$$B_{zi}(z) = \frac{2}{3}\mu_0 M$$

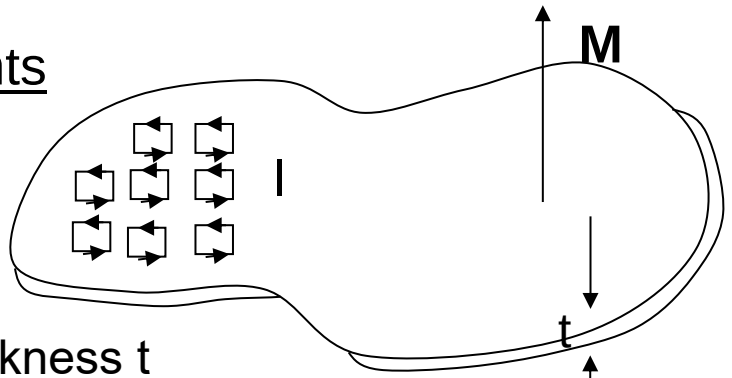
which a constant

similar to a spinning spherical shell

Outside the sphere, B is similar to a pure dipole of moment

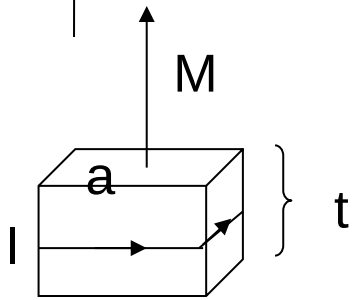
$$m = \frac{4}{3}\pi a^3 M$$

Physical interpretation of bound currents



Consider a thin slab of magnetized material with thin dipoles due to tiny loops

Consider a tiny loop with area ‘a’ and thickness t



In terms of magnetization M, the dipole moment m is

$$m = Mat$$

In terms of circulating current I, the dipole moment m is

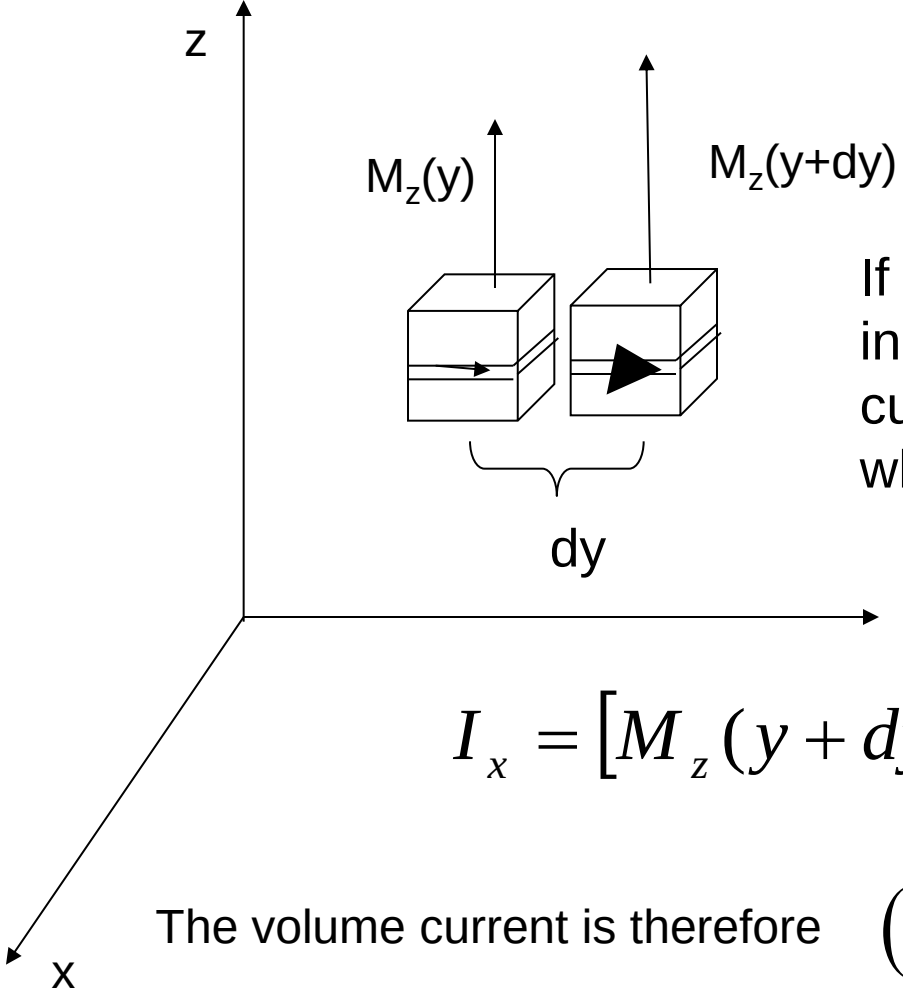
$$m = Ia$$

From the above two eqns. $I = Mt$

So the surface current is $K_b = I/t = M$

To be precise $\vec{K}_b = \vec{M} \times \hat{n}$ Where ň is normal to surface

No current at the top and bottom surface since M is parallel to n and the cross product vanishes. Current will be only along rim.



If there is Non-uniform magnetization in the z-direction, there will be a net current in the x-direction on the surface where they join

$$I_x = \left[M_z(y+dy) - M_z(y)\right]dz = \frac{\partial M_z}{\partial y}\,dydz$$

The volume current is therefore $\left(J_b\right)_x = \frac{\partial M_z}{\partial y}$

Similarly non-uniform magnetization in the y-direction will contribute $\left(J_b\right)_x = -\frac{\partial M_y}{\partial z}$

Hence $$\left(J_b\right)_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}$$

Taking contributions from all directions

$$\vec{J}_b = \vec{\nabla} \times \vec{M}$$

Like any other steady current

$$\vec{\nabla} \cdot \vec{J}_b = 0$$ since div curl is zero

The Auxiliary Field H

Effect of magnetization is to establish bound currents J_b within the material and K_b on the surface. The mag field is due to these bound currents. We say that the field is due to J_b plus all others which we call free current J_f

Total current $\vec{J} = \vec{J}_b + \vec{J}_f$

Ampere's law in terms of **B** is

$$\frac{1}{\mu_0}\left(\vec{\nabla}\times\vec{B}\right) = \vec{J} = \vec{J}_b + \vec{J}_f = \vec{J}_f + \left(\vec{\nabla}\times\vec{M}\right)$$

$$\vec{\nabla}\times\left(\frac{1}{\mu_0}\vec{B} - \vec{M}\right) = \vec{J}_f$$

$\bar{D} = \varepsilon_0\bar{E} + \bar{P}$

$\bar{H} = \bar{B}/\mu_0 - \bar{M}$

The term in brackets is called **H**

$$\vec{H} = \frac{1}{\mu_0}\vec{B} - \vec{M} \qquad \vec{\nabla}\times\vec{H} = \vec{J}_f$$

Ampere's law in terms of **H**

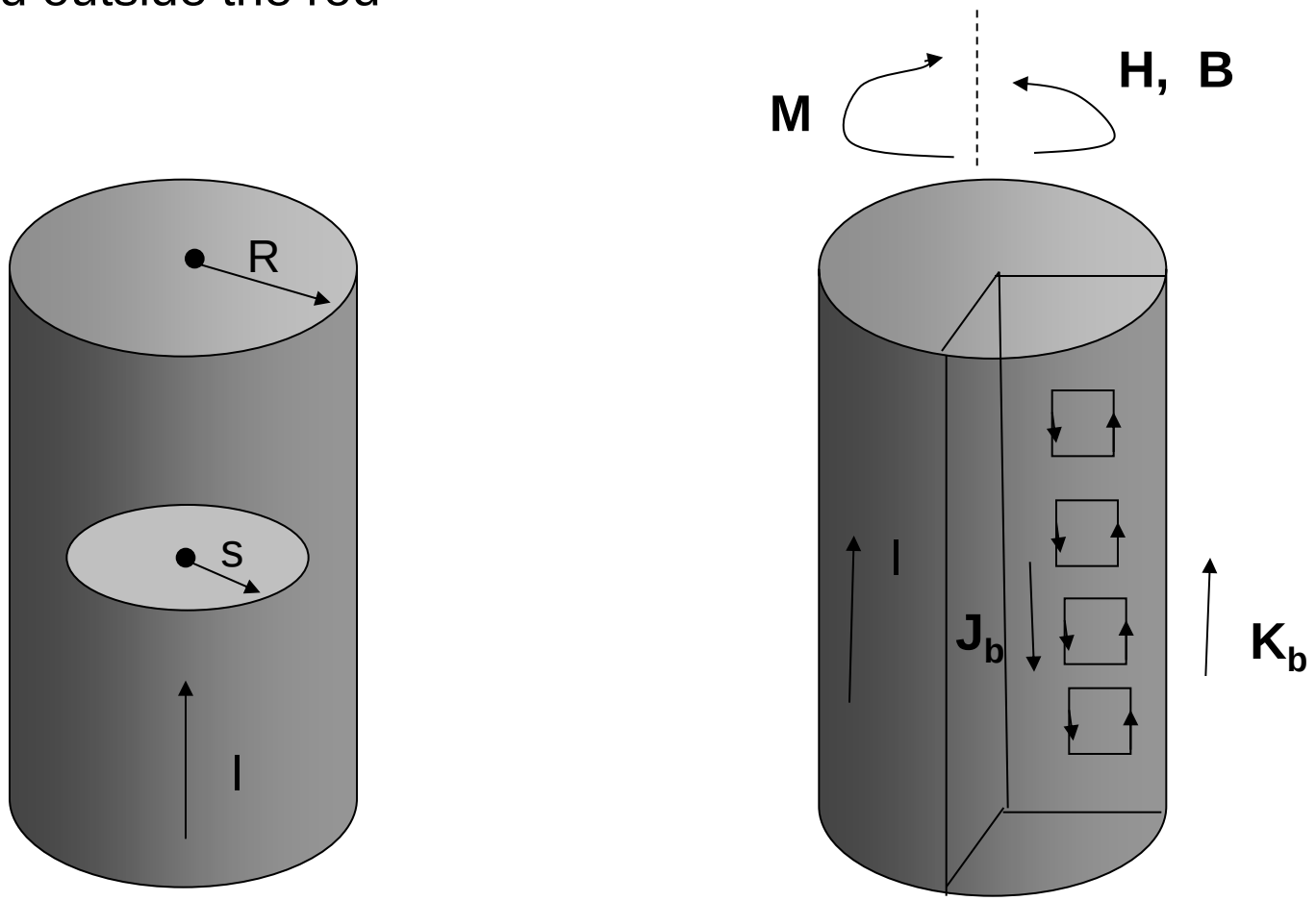
$\bar{\nabla}\cdot\bar{D} = \rho_f$

$\bar{\nabla}\times\bar{H} = \bar{J}_f$

In the integral form $\oint \vec{H}\cdot d\vec{l} = I_{f_{enc}}$

Example

A long copper rod of radius R carries a uniformly distributed (free) current I. Find H inside and outside the rod



Copper is weakly diamagnetic, hence the magnetization will be Opposite to the field, and the bound current will be antiparallel to I within, and parallel along the surface.

For s < R

$$H(2\pi s) = I_{f_{enc}} = I\frac{\pi s^2}{\pi R^2}$$

$$\vec{H} = \frac{I}{2\pi R^2} s\hat{e}_\phi$$

For s ≥ R

$$\vec{H} = \frac{I}{2\pi s}\hat{e}_\phi$$

Outside **M** = 0

$$\vec{B} = \mu_0\vec{H} = \frac{\mu_0 I}{2\pi s}\hat{e}_\phi$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

Divergence of H is in general not zero

$$\vec{\nabla} \cdot \vec{H} = -\vec{\nabla} \cdot \vec{M}$$

Boundary conditions

$$\vec{\nabla} \times \vec{H} = \vec{J}_f \qquad \vec{\nabla} \cdot \vec{H} = -\vec{\nabla} \cdot \vec{M}$$

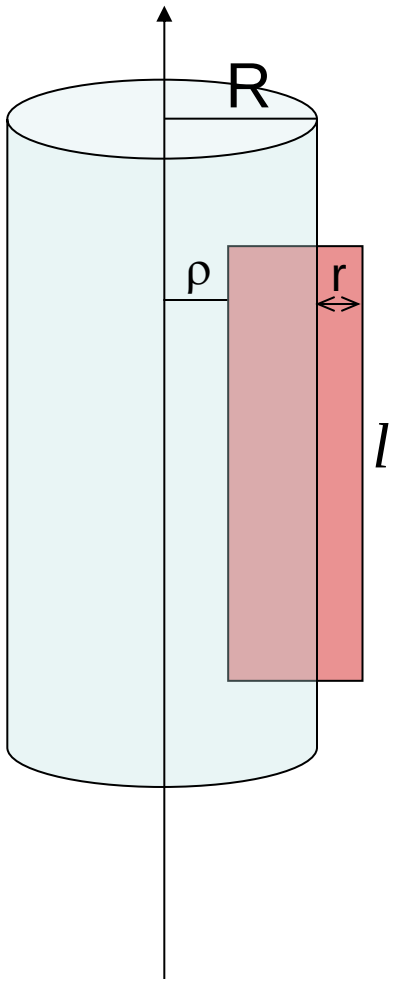
$$H^{\perp}_{above} - H^{\perp}_{below} = -\left(M^{\perp}_{above} - M^{\perp}_{below}\right)$$

$$\vec{H}^{\parallel}_{above} - \vec{H}^{\parallel}_{below} = \vec{K}_f \times \hat{n}$$

Compare with $B^{\perp}_{above} - B^{\perp}_{below} = 0$

$$\vec{B}^{\parallel}_{above} - \vec{B}^{\parallel}_{below} = \mu_0 \left(\vec{K}_f \times \hat{n}\right)$$

Calculate the magnetic field inside and outside a cylinder having a magnetization $\vec{M} = k\rho\hat{e}_z$ parallel to the axis (frozen in).



Now

$$\vec{J}_b = \nabla \times \vec{M} = -k\hat{e}_\varphi \text{ and } \vec{K}_b = \vec{M} \times \hat{e}_\rho = k\,\mathrm{R}\hat{\mathrm{e}}_\varphi$$

To calculate the magnetic field inside the cylinder, construct an Amperian loop as shown. Then,

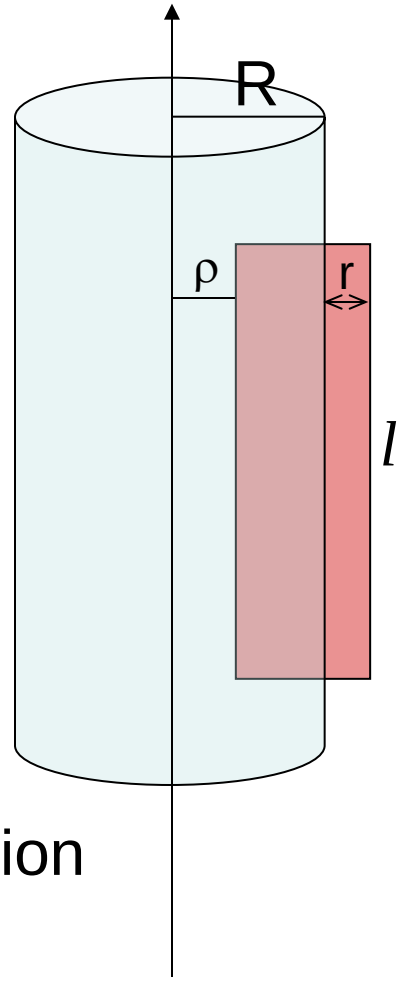
$$\oint \vec{B}.d\vec{l} = \mu_0 I_{enc} = \mu_0 \left[\int J_b da + \int K_b dl \right]$$

$$Bl = \mu_0 \left[-kl\left(R - \rho\right) + klR \right] = \mu_0 kl\rho$$

as the bound currents do not exist outside the cylinder.

$$Bl = \mu_0 \left[-kl\left(R - \rho\right) + klR \right] = \mu_0 kl\rho$$

$$\vec{B} = \mu_0 k \rho \hat{e}_z$$



The field is zero outside the cylinder.

By symmetry, the auxiliary field is in the $\hat{e}_z$ direction

$$\oint Hdl = \mu_0 I_{fenc} = 0$$

As there is no free current. Thus, the auxiliary field is zero.

Linear and Nonlinear media

Magnetic susceptibility and Permeability

For most substances, the magnetization is proportional to the field

$$\vec{M} = \chi_m \vec{H}$$ Materials that obey this are linear media

χ_m is the magnetic susceptibility is positive for paramagnets and negative for diamagnets

Now

$$\vec{B} = \mu_0\left(\vec{H} + \vec{M}\right) = \mu_0\left(1 + \chi_m\right)\vec{H} = \mu\vec{H}$$

where

$$\mu = \mu_0\left(1 + \chi_m\right)$$ is called the permeability of the material

For vacuum, there is no matter to magnetize, so susceptibility is zero

Hence $\mu = \mu_0$ is the permeability of free space